

3(a)	force $\times$ distance	M1
	perpendicular distance of (line of action of) force from the point	A1
3(b)(i)	distance moved by pointer = $123 - 86 (= 37 \text{ mm})$	C1
	(extension =) $37 \times (1.8 / 52.6) = 1.3 \text{ (mm)}$	A1
	or sin or tan $\theta = 37 / 526$ (so $\theta = 4.0^\circ$ so extension =) sin or tan $\theta \times 18 = 1.3 \text{ (mm)}$	
3(b)(ii)	moment = $0.472 \times 9.81 \times 6.2 \times 10^{-2}$	C1
	= 0.29 N m	A1
3(b)(iii)	$(\Delta)F \times 1.8 \times 10^{-2} = 0.29$	C1
	$\Delta F = 16 \text{ N}$	A1
3(b)(iv)	$k = F / x$	C1
	= $16 / (1.3 \times 10^{-3})$	A1
	= $1.2 \times 10^4 \text{ N m}^{-1}$	